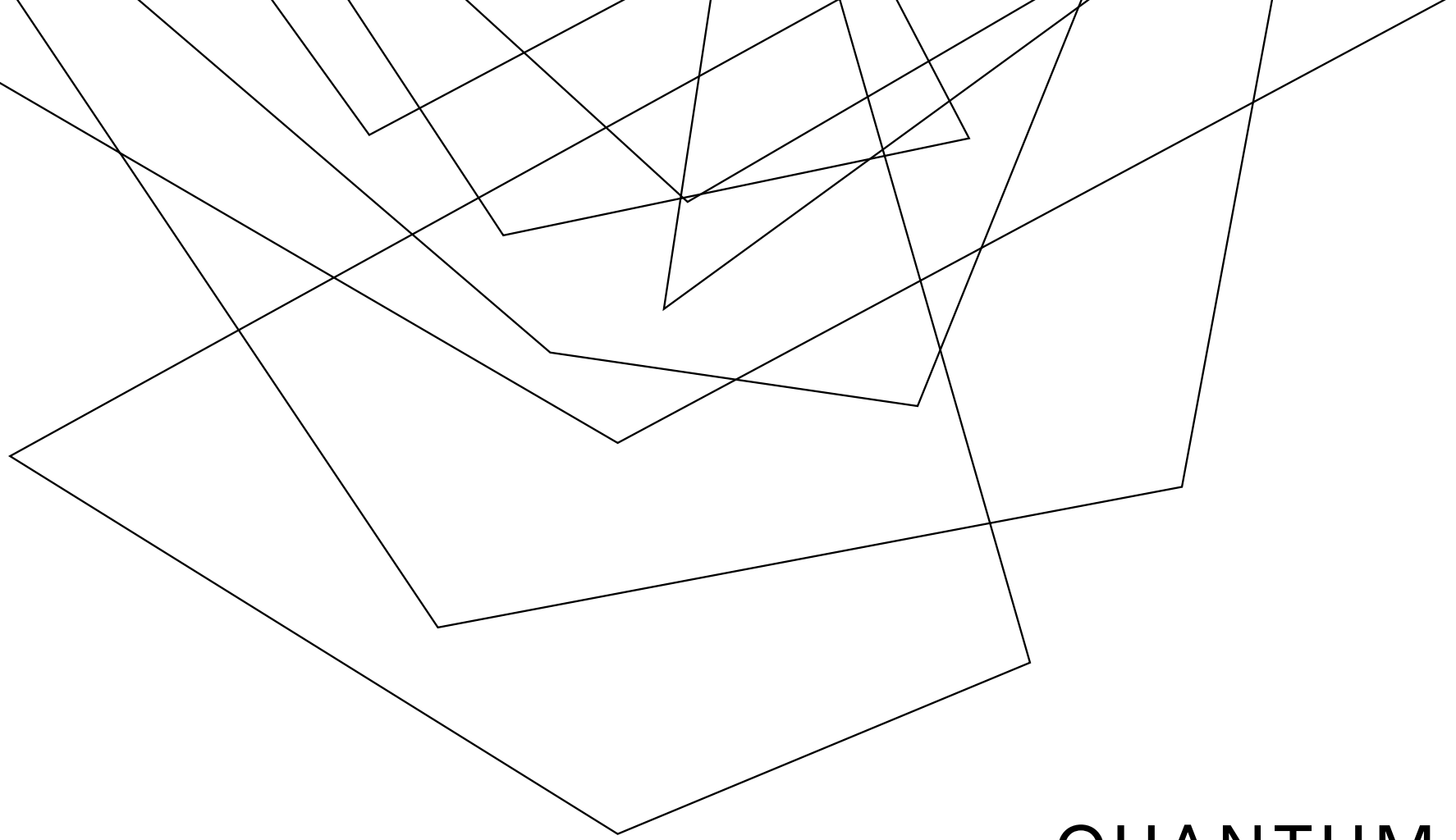


QUANTUM BREAK-IN

Nestors Starostins, ns23044

DESCRIPTION

A puzzle game.

The player needs to enter the password (target state) by applying quantum gates on qubits, in a limited number of steps.

There are sentinels who observe certain qubits at certain steps and raise alarm if they measure 1.

CURRENT STATE

Features & Gameplay

- Core concepts implemented (qubits, turns, sentinels, levels)
- 6 levels from tutorial to challenge
- Level design as JSON, which can specify initial states, targets, gate sets, step counts, sentinel behaviour

Quantum Concepts & Mechanics

- Quantum computing simulated with real numbers and arrays
- 7 built-in gates, easily extendable for more
- Superposition and entanglement
- Visualisation of quantum state through colour

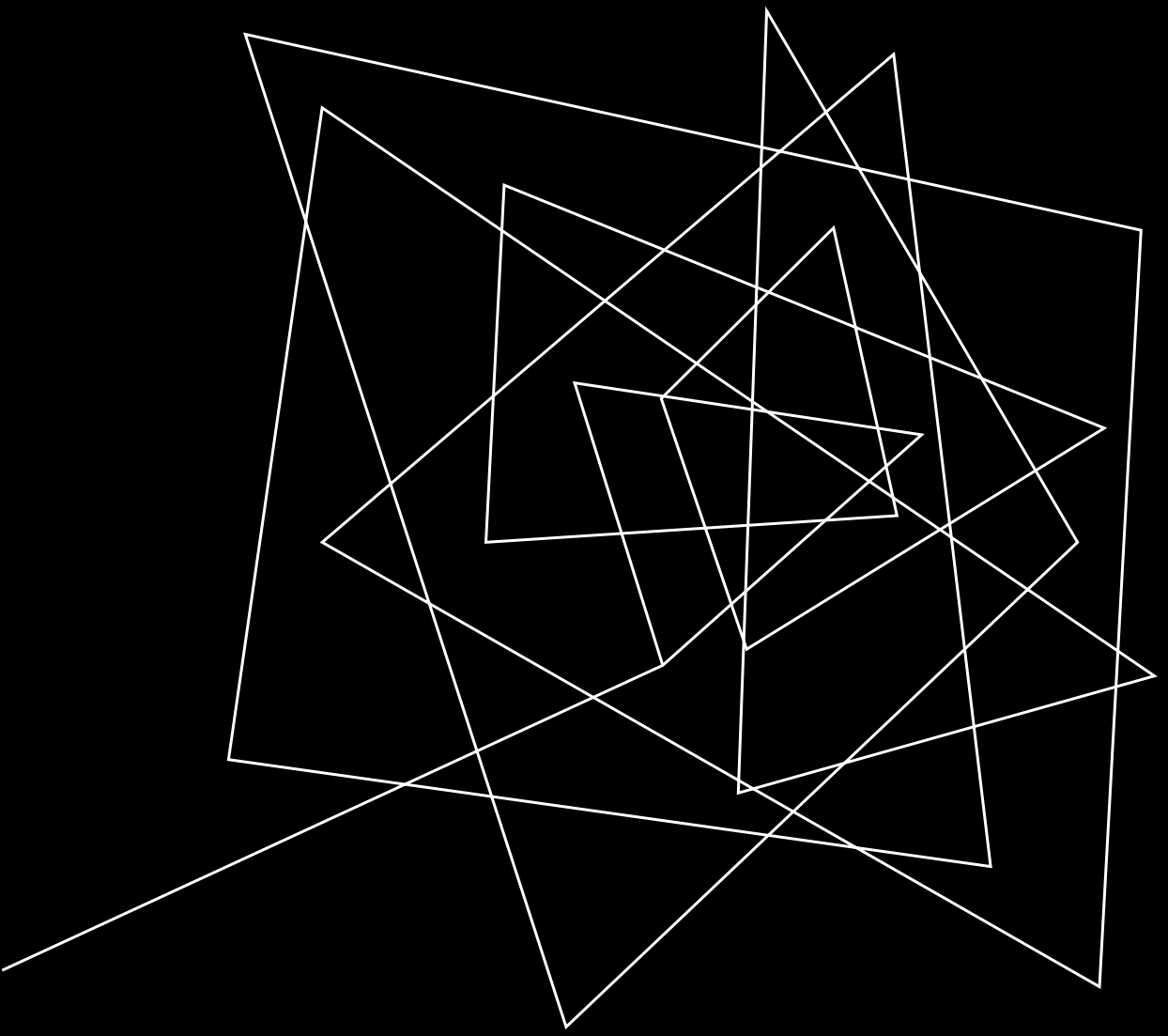
FURTHER PLANS

- Graphics, animations, sound effects
- Qubit grid and grid-dependent C-gates
- More levels and a tutorial

AI USE

Claude Sonnet 5.5 (free)

- Explained the basics of Godot engine
- Structured the project and development schedule
- Generated most of the project after prompted to make just the Quantum simulator part



THANK YOU!